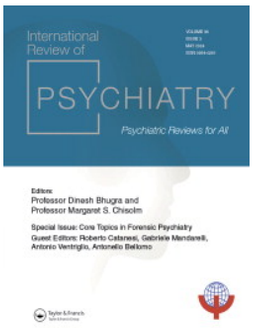

International Review of Psychiatry

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/iirp20

Repetitive head-injury exposure and later-in-life cognitive and emotional outcomes among former collegiate football players: a CLEAATS investigation

Jeff Schaffert, Alison Datoc, Gavin D. Sanders, Nyaz Didehbani, Christian LoBue & C. Munro Cullum

To cite this article: Jeff Schaffert, Alison Datoc, Gavin D. Sanders, Nyaz Didehbani, Christian LoBue & C. Munro Cullum (2024) Repetitive head-injury exposure and later-in-life cognitive and emotional outcomes among former collegiate football players: a CLEAATS investigation, International Review of Psychiatry, 36:3, 233-242, DOI: 10.1080/09540261.2024.2352572

To link to this article: https://doi.org/10.1080/09540261.2024.2352572

Published online: 14 May 2024.

Submit your article to this journal

Article views: 122

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=iirp20

INTERNATIONAL REVIEW OF PSYCHIATRY
2024, VOL. 36, NO. 3, 233–242
https://doi.org/10.1080/09540261.2024.2352572

Taylor & Francis
Taylor & Francis Group

RESEARCH ARTICLE

Repetitive head-injury exposure and later-in-life cognitive and emotional outcomes among former collegiate football players: a CLEAATS investigation

Jeff Schaffert[a], Alison Datoc[b], Gavin D. Sanders[c], Nyaz Didehbani[d], Christian LoBue[e] and C. Munro Cullum[f]

[a]Division of Psychology, University of Texas Southwestern Medical Center, Dallas, Texas, USA; [b]Department of Psychiatry, Children's Health Andrews Institute, Orthopaedics and Sports Medicine, Division of Psychology, University of Texas Southwestern Medical Center, Dallas, Texas, USA; [c]Department of Psychiatry, Division of Psychology, University of Texas Southwestern Medical Center, Dallas, Texas, USA; [d]Department of Psychiatry, Physical Medicine and Rehabilitation, Division of Psychology, University of Texas Southwestern Medical Center, Dallas, Texas, USA; [e]Department of Psychiatry, Neurological Surgery, Division of Psychology, University of Texas Southwestern Medical Center, Dallas, Texas, USA; [f]Department of Psychiatry, Neurology, and Neurological Surgery, Division of Psychology, University of Texas Southwestern Medical Center, Dallas, Texas, USA

ABSTRACT

This study measured the relationship between head-injury exposure and later-in-life cognitive and emotional symptoms in aging collegiate football players who participated in the College Level Aging Athlete Study. Linear regressions examined the relationship between various head-injury exposure variables (head-injury exposure estimate [HIEE], number of diagnosed concussions, and symptomatic hits to the head) and subjective cognitive function, objective cognitive function, and emotional/mood symptoms. Additional regressions evaluated the impact of emotional symptoms on subjective cognitive decline and objective cognitive function. Participants (n = 216) were 50–87 years old (M = 63.4 [8.5]), 91% White, and well-educated (bachelor's/graduate degree = 92%). HIEE did not predict scores on cognitive or emotional/mood symptom measures (*p's* > .169). Diagnosed concussions had a small effect on depression symptoms (p = .002, b = 0.501, R2 = .052) and subjective cognitive symptoms (p = .002, b = 0.383, R2 = .051). An emotional symptom index had a stronger relationship (p < .001, b = 0.693, R2 = .362) with subjective cognitive functioning but no significant relationship with objective cognitive function (p = .052, b = −0.211, R2 = .020). Controlling for emotional symptoms, the relationship between concussions and subjective cognitive symptoms was attenuated (p = .078, R2 = .011). Findings suggested that head-injury exposure was not significantly related to cognitive or emotional/mood outcomes in former collegiate football players and highlighted the importance of current emotional/mood symptoms on subjective cognitive function.

ARTICLE HISTORY
Received 2 April 2024
Accepted 2 May 2024

KEYWORDS
Amateur sports; concussions; cognitive decline; repetitive head-injury

Introduction

Participation in football has been associated with increased risk of dementia and emotional difficulties later-in-life (Mez et al., 2020), with some evidence of a dose-response relationship between repetitive head-injury (RHI) exposure and later neurocognitive and emotional difficulties (Alosco et al., 2017; Stamm et al., 2015). This has led to widespread interest among the general population, researchers, athletes, and media regarding the risk of dementia among football players. However, scientific findings regarding the dose-response relationship of RHI exposure and poor cognitive and emotional outcomes have been mixed among retired National Football League (NFL) players. Some studies have shown that retired NFL players with increasing RHI exposure have higher risk of depression (Guskiewicz et al., 2007). Others have found higher rates of mild cognitive impairment (Randolph et al., 2013), and dementia (Lehman et al., 2012) compared to matched cohorts, while other investigations have failed to demonstrate a consistent dose-response relationship between increased RHI exposure and cognitive outcomes (Fields et al., 2019; Schaffert et al., 2019; 2022). However, much of the

CONTACT Jeff Schaffert jeffrey.schaffert@utsouthwestern.edu Division of Psychology, University of Texas Southwestern Medical Center, 5323 Harry Hines Blvd., Dallas, TX 75390-9044, USA.
This article has been corrected with minor changes. These changes do not impact the academic content of the article.
© 2024 University of Texas Southwestern Medical Center

current literature investigating long-term cognitive and emotional outcomes following increased RHI exposure has targeted retired professional football players, and it is unclear if findings at the professional level translate to amateur football participants who are more numerous and arguably have less RHI exposure than players at the professional level.

Among American style football players at amateur levels, most studies evaluating later-in-life outcomes have examined high school football athletes, with a recent systematic review finding that no studies to date on former high school football players have shown a link between participation in contact sport at this level and later-in-life mental health or cognitive problems (Iverson et al., 2023). Studies that have evaluated a dose-response relationship of RHI exposure (such as age at first exposure in contact sports) on later-in-life cognitive or emotional symptoms in former amateur athletes have yielded conflicting findings. Most studies have failed to find a link between age of first exposure to football and poor cognitive or emotional outcomes (Brett et al., 2019; Caccese et al., 2019; 2020; 2020; Iverson et al., 2021; 2021; 2021), though there is some evidence that cumulative head-injury exposure over time may have a relationship with brain health later-in-life among former amateur contact sport athletes. In the LEGENDS cohort out of Boston University, a cumulative head-injury exposure index was predictive of risk of cognitive impairment, apathy, depression, and behavioral dysregulation in 93 former high school (n = 17) and college football players (n = 76;Montenigro et al., 2017). A more recent study in 2022 examining 58 former NCAA football players found that a Head Impact Exposure Estimate (HIEE), but not years of play, was predictive of worse delayed recall on a measure of verbal memory, as well as greater self-reported difficulties with cognition and overall distress (Brett et al., 2022). These findings suggest that proxy variables for head-injury exposure (such as years of play or age of first exposure versus a more objective cumulative head-injury estimate) may fail to detect the impact of RHI exposure on later-in-life cognitive function, and raise questions regarding the best way to measure head-injury exposure and the potential negative outcomes related to contact sports participation.

In addition to investigating the associations between estimated head injury exposure and objective cognitive function later-in-life, researchers have also investigated subjective cognitive difficulties in former athletes (Bryant et al., 2023; Stillman et al., 2020). Subjective cognitive concerns or subjective cognitive difficulties (we will use these terms interchangeably) refers to a person's own concerns or perceptions regarding their cognitive abilities, regardless of their actual abilities on cognitive screening measures or neuropsychological assessment (i.e. objective cognitive performance). Subjective cognitive difficulties have not always shown strong relationships with objective cognitive performance in cognitively normal populations (Reid et al., 2012) or athletes after a concussion (Bryant et al., 2023). Instead, greater subjective symptoms have been more consistently linked to increased depressive and anxious symptoms (Brett et al., 2023; Bryant et al., 2023; Reid et al., 2012; Stillman et al., 2020), which can consequently negatively impact objective cognitive function. As such, subjective cognitive symptoms alone may not be adequate indicators of later-in-life cognitive functioning in former athletes, and should be considered in the context of objective cognitive data and other potential factors known to impact cognition such as mood disturbance.

Currently, the scientific community has no consensus on the amount of RHI exposure that is needed to increase the risk of mental health or cognitive difficulties in former professional or amateur athletes, or identified the most important factors that influence emotional or cognitive outcomes. Understanding the long-term impact of RHI exposure on later-in-life brain health is paramount, not only for athletes themselves, but also for medical professionals and coaches to ensure the safety and well-being of players, while considering the numerous health benefits derived from active involvement in sports. Participation in sports is known to be associated with a myriad of health benefits, including physical activity, psychological well-being, and social connectedness (Eime et al., 2013; Malm et al., 2019; Mandolesi et al., 2018), all of which have been shown to be important predictors for brain health across the lifespan. As such, the relationship between RHI exposure and later-in-life brain health outcomes is complex and requires further study to weigh potential risks with the well-established health advantages for athletes at all participation levels.

Overall, there is a paucity of research on long-term cognitive and emotional outcomes in aging amateur athletes, and the relationship between RHI exposure and long-term outcomes in these athletes remains unclear. The aim of this investigation was to examine the relationship between RHI exposure variables (i.e. a head-injury exposure index, number of diagnosed concussions, and number of symptomatic hits to the head) and later-in-life objective cognitive functioning, subjective cognitive concerns, and emotional symptoms (i.e. depression, anxiety, and emotional and behavioral dysregulation) among former collegiate football players. We elected to investigate various measures of RHI exposure, including a cumulative head-injury index score that was independent of

number of concussions given past research that has shown a relationship with cognitive and emotional outcomes (Brett et al., 2022; Montenigro et al., 2017). Given past research in the area, we hypothesized that RHI exposure variables would be weakly associated with objective cognitive performance and anticipated that subjective cognitive difficulties were going to positively correlated with emotional symptoms such as anxiety and depression.

Methods

Participants

Participants in this investigation came from the Collegiate Level Aging Athlete Study (CLEAATS), an ongoing study launched in 2021. The CLEAATS study includes brain health outcome from survey and cognitive data on over 650 male and female athletes aged 50+.

Inclusion criteria for the current study included former male collegiate football players with at least one season of football in a National Collegiate Athletics Association (NCAA) or National Association of Intercollegiate Athletics (NAIA) school, and having complete data on high school and collegiate positions, number of games played, and the average estimated percentage of time played in each game to calculate the head injury exposure index (HIEI). All participants consented and agreed to participate in the study, which followed Helsinki Declaration guidelines and was approved by the University of Texas Southwest Medical Center International Review Board (STU-2021-1101).

Measures

Head-injury exposure index

A HIEI estimates cumulative head injury exposure and has been reported to be sensitive in predicting cognitive and emotional outcomes in younger amateur cohorts (Montenigro et al., 2017) and retired professional football players (Wright et al., 2016). A HIEI in this study was calculated based on a modified version published in 2015 by Kerr and colleagues. The cumulative HIEI was based on total high school and college season contact practices per week and their length of time, games played, and position played (including one-way and two-way players). Weights of head-injury exposure were provided for each position played in high school and college based on published helmet accelerometer data (see Kerr et al., 2015 for more details). This was a modified version of the original published index given that Kerr and colleagues included data from pre-season and post-season data (Kerr et al., 2015). CLEAATS did not separate this data and only asked respondents to report games played, positions played, number of contact practices per week, and average length of contact practices. As such, these data were used to calculate the HIEI in the current study, and a summation of games played, number of contact practice sessions (plus duration per week), and time spent in game (i.e. one-way versus two-way player), weighted by primary position played. A head-injury exposure estimate was created for each season of high school and college football, and a total sum was derived for each participant. See Table 1 for an example on how the modified HIEI was calculated. This example consists of a 4-year high school and college player who switched positions in high school (and played both offensive and defensive positions).

Self-reported diagnosed concussions and symptomatic hits

CLEAATS participants also provide a detailed history on diagnosed and possible concussions. For the

Table 1. Example of head-injury exposure index calculation.

	High School				College				
	1	2	3	4	1	2	3	4	Overall
Primary Position	Linebacker	O-Line	O-Line	O-Line	O-Line	O-Line	O-Line	O-Line	
Secondary Position	O-Line	D-Line	D-Line	D-Line	N/A	N/A	N/A	N/A	
Contact Practices/ Week	2	3	3	3	3	3	3	3	
Number of Contact Weeks	14	14	14	14	16	16	16	16	
Hours per Practice	0.75	1.5	1.5	1.5	1.0	1.0	1.0	1.0	
Total Practice Contact Hours	21	63	63	63	48	48	48	48	402
Number of Games	6	10	10	12	10	10	10	12	
% of Active Time	25%	75%	25%	25%	50%	50%	50%	50%	
Time within Game	1.00	1.00	1.00	1.00	0.50	0.50	0.50	0.50	
Position Weight (Frequency & Severity)	3.3×1.06	2.68×1.03	2.68×1.03	2.68×1.03	2.19×0.98	2.19×0.98	2.19×0.98	2.19×0.98	
Total Game Contact Hours	5.25	20.70	6.90	8.28	5.37	5.366	5.37	6.44	63.68
Total Head-Injury Exposure Index	26.25	83.70	69.90	71.28	53.37	53.37	53.37	54.44	465.68

*For players who play offense or defense (i.e. no secondary position) = 0.50; for players who play both offense and defense within a game = 1.00; for players who play a primary offensive or defensive positions and special teams = 0.55. CLEAATS did not collect data on number of contact weeks so this was held constant for each participant at 14 contact sessions per season for high school and 16 contact sessions per season for college.

number of self-reported diagnosed concussions, participants were asked to report the number of diagnosed concussions and how many of those resulted in loss of consciousness. Participants also reported possible concussions that were either diagnosed or went undiagnosed by asking them to estimate the "number of times they hit their head and noticed the following symptoms: being knocked out or unconscious, headache/migraine, dizziness, feeling lightheaded or dizzy, feeling sick or throwing up, feeling unsteady or off-balance, ringing in ears, changes in vision or eyesight, difficulty sleeping, seizures, lightheadedness, or feeling dazed or confused."

Cognitive functioning

Subjective cognitive functioning was defined using the Cognitive Functioning Instrument (CFI), a validated 14-item measure of subjective cognitive decline. The CFI measures cognitive symptoms over the last year and is commonly used in aging dementia populations, where higher scores indicate more subjective cognitive decline from a scale of 0 to 14 (Amariglio et al., 2015; Ferris et al., 2006; Li et al., 2017). Objective cognitive functioning was measured using the modified-Telephone Interview for Cognitive Status (TICS-m), a 50-item telephone administered cognitive screening that has been well-researched (Carlew et al., 2020). Higher scores on the TICS-m indicate better cognitive functioning. In this study, subjective cognitive functioning and objective cognitive functioning were treated as separate constructs (i.e. presence of subjective cognitive concerns did not depend on the absence of objective cognitive impairment, and participants could have both subjective concerns and objective cognitive impairment).

Emotional functioning

Three primary measures were utilized in this study to assess emotional functioning. Depressive symptoms were measured using the Patient Health Questionnaire (PHQ; 8-item version). Anxiety symptoms were measured using the Generalized Anxiety Disorder Questionnaire (GAD-7). Both the PHQ and GAD-7 have been well validated as screening tools for depression and anxiety in a variety of populations (Kroenke et al., 2001; 2009; Spitzer et al., 2006). Emotional dysregulation was assessed using the Neurological Quality of Life Emotional and Behavioral Dyscontrol (NQoL EBD), which is an 18-item scale measuring disinhibition, emotional lability, irritability, impatience, impulsiveness, and explosivity (Carlozzi et al., 2020; Cella et al., 2012; Gershon et al., 2012).

Analyses

Hierarchical multiple linear regressions adjusting for age and education in the first block were used to evaluate the impact of the HIEI on the subjective cognitive functioning (CFI), objective cognitive functioning (TICS-m), and emotional (PHQ-8, GAD-7, NQoL EBD) measures. We chose to enter these data in separate blocks to measure the added impact of RHI exposure variables to our cognitive and emotional variables. Additional hierarchical multiple linear regressions (controlling for age) were also conducted to assess for the influence that emotional symptoms (i.e. PHQ-8, GAD-7, NQoL EBD) may have with subjective and objective cognitive functioning (i.e. CFI and TICS-m). For these additional analyses, we utilized an emotional index score (a centered Z-score average of PHQ-8, GAD-7, and NQoL EBD) to create more parsimonious model, rather than entering each of these variables separately. P-values were set at a more conservative $p < .01$ to reduce the chances of type-one error given the number of analyses conducted. Missing data was handled list-wise (i.e. analysis by analysis).

Results

Sample characteristics

Participant demographics and head-injury exposure variables can be found in Table 2. Out of 243 total football players, 216 completed survey questions that allowed calculation of the HIEI, and 157 (73%) completed the TICS-m in a follow-up phone call. Former players were on average 64.1 years old (range = 50 to 87), were mostly White (92%), well-educated (55% with a graduate degree), and played contact football for an average of 10.5 years. Approximately 20% of the sample reported a clinician diagnosis of mild cognitive impairment at the time of the survey (n = 44) with an average age of cognitive symptom onset of 56.2 years (SD = 11.5; Min = 30; Max = 75), and 59% (n = 25) reported a progressive decline over time. Three clear outliers were removed from the symptomatic hits/possible concussion analyses, as two participants reported 100 symptomatic hits and one participant reported 300 symptomatic hits.

Multiple linear regressions: repetitive head-injury exposure and cognitive and emotional symptoms

Results from primary multiple linear regressions can be found in Table 3. Results suggested that the HIEI did not predict TICS-m, CFI, PHQ-8, GAD-7, or

NQoL EBD scores. Self-reported symptomatic hits did not significantly relate to the TICS-m, CFI, PHQ-8, or GAD-7 scores, but did significantly predict self-reported emotional and behavioral dysregulation (i.e. NQoL EBD). However, overall effect size was small, with only a 0.249 unstandardized beta-weight (i.e. each additional hit correlated with a 0.249 increase in NQoL EBD total score) and 4.2% of variance explained. Lastly, diagnosed concussions did not significantly predict TICS-m, GAD-7, or NQoL EBD, though it was significantly related to PHQ-8 (~5% of explained variance) and CFI (~5% of variance explained) with a fairly small unstandardized beta weight, with each reported concussion predicting ~0.5 point increase on the PHQ-8 and ~0.4 point increase on the CFI. While not meeting the current cut-off for statistical significance (p = .012) each diagnosed concussion predicted approximately a 1-point increase on the NQoL EBD.

Multiple linear regressions: emotional functioning and cognitive symptoms

We performed two additional multiple linear regressions, with age and education entered in the first block, and an emotional index score (Z-score mean of PHQ-8, GAD-7, and NQoL EBD) entered into the second block, predicting CFI and TICS-m in separate regressions. Results can be found in Table 4 below. Emotional index correlated strongly (r = .581) with CFI scores (i.e. subjective report of cognitive symptoms), and explained 36% of the variance in report of symptoms on the CFI. Emotional functioning did not correlate with objective cognitive performance (i.e. TICS-m, r = −.082) and was not a significant predictor of TICS-m scores.

Table 2. Demographics, football and head-injury exposure, and cognitive and emotional functioning.

	M or n	SD or %	Min.	Max
Age (n = 216)	64.1	8.5	50	87
Race/Ethnicity (n = 216)				
White	199	92		
Black	4	2		
Hispanic	6	3		
American Indian	3	1		
Multi-Racial	4	2		
Education (n = 216)				
Some College No Degree	9	4		
Associates Degree	2	1		
Bachelor's Degree	85	39		
Master's Degree	75	35		
Doctoral Degree	45	21		
Total Years of Football (n = 215)	10.5	3.2	1	22
Age Beginning Football (n = 216)	11.0	2.4	5	17
Years Since Play (n = 215)	42.5	8.8	26	66
Total Head-Injury Exposure Index (n = 216)	763.8	342.9	6.4	1839.5
Possible Concussions/ Symptomatic Hits (n = 189)	6.9	8.7	0	60
Diagnosed Concussions (n = 192)	1.3	1.9	0	10
Cognitive Functioning Index (n = 209)	3.3	3.1	0	14
TICS-m (n = 157)	38.3	4.1	20	50
PHQ-8 (n = 206)	3.0	4.0	0	22
GAD-7 (n = 206)	2.1	3.4	0	21
NQoL EBD (n = 205)	27.6	10.5	18	72

Note: TICS-m = Modified Telephone Interview for Cognitive Status; PHQ-8 = Patient Health Questionnaire (suicide item omitted); GAD-7 = Generalized Anxiety Disorder Scale; NQoL EBD = Neurological Quality of Life Emotional and Behavioral Dyscontrol Scale.

Post-hoc analysis

Given the significant finding between number of concussions and subjective cognitive functioning. An additional post-hoc multiple linear regression examining the relationship between these variables was conducted controlling for emotional symptoms. In this analysis, age, education, and the mean Z-score of PHQ-8, GAD-7, and NQoL EBD were included in the first block and diagnosed concussions were in the second block. When controlling for emotional symptoms, the relationship between number of concussions and CFI scores was attenuated (F = 3.239, p = .078, R2 = .011).

Discussion

There are generally equivocal findings in the literature regarding the long-term consequences of sports-related RHI exposure in amateur athletes on cognitive and psychiatric health. In the present study, various estimates of RHI exposure (i.e. cumulative exposure, number of self-reported diagnosed concussion and symptomatic hits) were differentially related to subjective cognitive decline and emotional symptoms but were not predictive of objective cognitive functioning. Increased emotional symptoms were strongly associated with subjective cognitive decline, whereas there was not a significant relationship between emotional distress and objective cognition. Lastly, the relationship between head-injury exposure (i.e. diagnosed concussions) and subjective cognitive report was attenuated when controlling for emotional symptoms. Findings from this study elucidate the complex relationship among these factors and offer little support to suggest that RHI exposure in former amateur athletes is specifically associated with negative cognitive outcomes.

There is a high level of concern in the public regarding the potential impact of RHI exposure on long-term brain health. Though some studies have reported a link between increased RHI exposure (as

Table 3. Multiple linear regressions controlling for age and education.

	F-change	*p*	Unstandardized beta (95% CI)	R^2 - Change
CFI (n = 209)				
Block 1: Age + Education	0.860	.425		
Block 2: Age + Education + HIEI	0.030	.862	0.000 (−0.001 to 0.001)	.000
TICS-m (n = 157)				
Block 1: Age + Education	18.251	< .001*		
Block 2: Age + Education + HIEI	1.906	.169	0.001 (−0.001, 0.003)	.009
PHQ-8 (n = 206)				
Block 1: Age + Education	3.032	.050		
Block 2: Age + Education + HIEI	0.062	.804	0.000 (−0.001 to 0.002)	.000
GAD-7 (n = 206)				
Block 1: Age + Education	1.638	.197		
Block 2: Age + Education + HIEI	0.000	.998	0.000 (−0.001, 0.001)	.000
NQoL EBD (n = 205)				
Block 1: Age + Education	1.717	.182		
Block 2: Age + Education + HIEI	0.003	.955	0.000 (−0.004, 0.004)	.000
CFI (n = 183)				
Block 1: Age + Education	1.042	.355		
Block 2: Age + Education + Symptomatic Hits	4.774	.030	0.056 (0.005, 0.106)	.025
TICS-m (n = 139)				
Block 1: Age + Education	17.318	< .001*		
Block 2: Age + Education + Symptomatic Hits	1.989	.161	−0.048 (−0.115, 0.019)	.012
PHQ-8 (n = 181)				
Block 1: Age + Education	1.894	.154		
Block 2: Age + Education + Symptomatic Hits	0.628	.429	0.025 (−0.038, 0.089)	.003
GAD-7 (n = 181)				
Block 1: Age + Education	1.180	.310		
Block 2: Age + Education + Symptomatic Hits	1.270	.286	0.036 (−0.023, 0.094)	.008
NQoL EBD (n = 180)				
Block 1: Age + Education	0.901	.408		
Block 2: Age + Education + Symptomatic Hits	7.861	.006*	0.249 (0.074, 0.425)	.042
CFI (n = 186)				
Block 1: Age + Education	1.042	.355		
Block 2: Age + Education + Diagnosed Concussions	9.981	.002*	0.383 (0.143, 0.623)	.051
TICS-m (n = 142)				
Block 1: Age + Education	17.315	< .001*		
Block 2: Age + Education + Diagnosed Concussions	0.389	.534	0.102 (−0.222, 0.426)	.002
PHQ-8 (n = 184)				
Block 1: Age + Education	2.483	.086		
Block 2: Age + Education + Diagnosed Concussions	10.111	.002*	0.501 (0.190–0.812)	.052
GAD-7 (n = 184)				
Block 1: Age + Education	1.644	.196		
Block 2: Age + Education + Diagnosed Concussions	4.336	.039	0.290 (0.015, 0.564)	.029
NQoL EBD (n = 183)				
Block 1: Age + Education	1.204	.302		
Block 2: Age + Education + Diagnosed Concussions	6.438	.012	1.067 (0.237, 1.896)	.034

Note: *p* – values denote "F-Change"; HIEI = Head-Injury Exposure Index; CFI = Cognitive Functioning Index; TICS-m = Modified Telephone Interview for Cognitive Status; PHQ-8 = Patient Health Questionnaire; GAD-7 = Generalized Anxiety Disorder Scale; NQoL EBD = Neurological Quality of Life Emotional and Behavioral Dyscontrol Scale. Different sample size in the analyses were the result of list-wise deletion secondary to some participants not completing portions of the survey.

Table 4. Multiple linear regressions of emotional functioning and cognitive symptoms.

	F-Change	*p*	Unstandardized Beta (95% CI)	R^2 - Change
CFI (n = 205)				
Block 1: Age + Education	1.040	.355		
Block 2: Age + Education + Emotional Index	115.347	<.001*	0.693 (0.566, 0.820)	.362
TICS-m (n = 157)				
Block 1: Age + Education	18.251	<.001*		
Block 2: Age + Education + Emotional Index	3.852	.052	−0.211 (-0.424, 0.001)	.020

measured through total index scores), lower cognitive performance, and greater psychological distress in former collegiate football players (Brett et al., 2022; Montenigro et al., 2017), the current results suggested a very subtle impact of total RHI exposure on only subjective cognitive concerns that appeared driven by current emotional symptoms. We did find a very small effect of number of concussions and symptomatic hits on depression and emotional dysregulation, respectively. However, the literature is at best inconclusive regarding the impact of concussion history on later-in-life emotional outcomes at the amateur football level, as while there has been an association between depression and concussions in professional players (Guskiewicz et al., 2007), these relationships have not been supported in amateur

athletes (Deshpande et al., 2017; McAllister & McCrea, 2017). In the present study, small effects (i.e. unstandardized beta-weights <= 0.5 points on the various measures) were observed between concussion history and current depression symptoms, as well as symptomatic hits and emotional and self-reported behavioral regulation. Given these small effects, the clinical significance of these findings is questionable. It is also important to note that given the strong relationship between emotional functioning and report of symptoms, the possibility that depression could impact the reported number of concussions or symptomatic hits cannot be excluded, as depression may heighten reports of negative prior health factors and has been shown to increase symptom reporting in the acute phase (Lange et al., 2011). As such, it is not possible to exclude the possibility of reverse causation in this sample, especially given the survey design of this study.

The most prominent finding in our study was that emotional symptoms had a strong correlation with subjective cognition (but not objective cognitive function) and the relationship between diagnosed concussions and subjective cognitive report was no longer significant when controlling for emotional factors. These results support existing evidence that have shown poorer subjective cognitive functioning related to psychological distress, as opposed to objective cognitive functioning (Bryant et al., 2023; Reid et al., 2012; Van Patten & Iverson, 2023). For example a recent paper found that among 57 former collegiate football players (15 years post-football participation), concussion history was inversely associated with performance on self-reported cognitive difficulties but not objective measures of cognition (Bryant et al., 2023). This aligns with our current findings and suggests that reliance on self-report of subjective cognitive functioning should not be used in isolation, but rather, considered with an individual's current emotional status. These data also have important clinical implications for sports neuropsychologists, as it strongly emphasizes the need for clinicians to consider the impact of current mood symptoms when evaluating players or former players with cognitive complaints.

Limitations to this study include reliance on self-report for history of diagnosed concussion and symptomatic hits, as well as recall of playing history to calculate a player's HIEI. Reliance on retrospective recall may be subject to bias and further research including a more objective measure of head injury exposure is warranted to be able to accurately quantify this information. Interestingly, the number of self-reported concussion diagnoses was low among the participants in this study, while possible concussions (symptomatic hits) were reported at a notably higher frequency. The discrepancy between these two variables is unclear, though it is possible that athletes in the sample experienced undiagnosed concussions. Second, the present study did not include other factors that may impact cognitive and emotional functioning such as medical history, and future research should examine lifestyle factors and medical conditions known to be risk factors for cognitive decline (e.g. exercise, substance use cardiovascular; sleep apnea) in these players. These results are also limited to a relatively homogeneous sample of former collegiate football players who were well-educated, White, and male. Future investigators should consider replicating these results to more diverse samples of former collegiate athletes who participated in sports outside of football, as well as athletes at other amateur participation levels. Finally, objective cognitive functioning was measured using a cognitive screening measure administered via telephone. The addition of more comprehensive neuropsychological testing would be beneficial in future studies to provide a more thorough understanding of former collegiate athletes' performance across cognitive domains, as it is possible that brief measures may fail to capture subtle declines in cognition.

In the present study of over 200 former collegiate football players, findings did not suggest a strong relationship between RHI exposure (measured in various ways) and cognitive and emotional outcomes later-in-life. Conversely, emotional symptoms were strongly related to self-reported subjective cognitive decline, indicating that emotional distress is a prominent contributor to former athletes' reports of cognitive decline. These results warrant caution in interpreting studies that do not measure cognitive decline objectively and rely on an athlete's report of symptoms (or next of kin in autopsy studies) without considering current emotional status. Future longitudinal investigations evaluating this complex relationship between RHI exposure and cognitive and emotional outcomes with comprehensive neurocognitive data are warranted.

Acknowledgements

This work was funded primarily by the Darrell K. Royal Research Fund, and supported in part by the Peter O'Donnell Jr. Brain Institute at UT Southwestern Medical Center.

Disclosure statement

Authors support no conflicts of interest pertinent to the current study.

ORCID

Jeff Schaffert http://orcid.org/0000-0002-8842-980X
Alison Datoc http://orcid.org/0000-0003-4464-2104
Gavin D. Sanders http://orcid.org/0000-0002-8231-8949
Nyaz Didehbani http://orcid.org/0000-0001-6121-5759
Christian LoBue http://orcid.org/0000-0001-9671-5526
C. Munro Cullum http://orcid.org/0000-0001-9706-5465

References

Alosco, M. L., Kasimis, A. B., Stamm, J. M., Chua, A. S., Baugh, C. M., Daneshvar, D. H., Robbins, C. A., Mariani, M., Hayden, J., Conneely, S., Au, R., Torres, A., McClean, M. D., McKee, A. C., Cantu, R. C., Mez, J., Nowinski, C. J., Martin, B. M., Chaisson, C. E., Tripodis, Y., & Stern, R. A. (2017). Age of first exposure to American football and long-term neuropsychiatric and cognitive outcomes. *Translational Psychiatry*, *7*(9), e1236–e1236. https://doi.org/10.1038/tp.2017.197

Amariglio, R. E., Donohue, M. C., Marshall, G. A., Rentz, D. M., Salmon, D. P., Ferris, S. H., Karantzoulis, S., Aisen, P. S., & Sperling, R. A. (2015). Tracking early decline in cognitive function in older individuals at risk for Alzheimer disease dementia: The Alzheimer's Disease Cooperative Study Cognitive Function Instrument. *JAMA Neurology*, *72*(4), 446–454. https://doi.org/10.1001/jamaneurol.2014.3375

Brett, B. L., Huber, D. L., Wild, A., Nelson, L. D., & McCrea, M. A. (2019). Age of first exposure to American football and behavioral, cognitive, psychological, and physical outcomes in high school and collegiate football players. *Sports Health*, *11*(4), 332–342. https://doi.org/10.1177/1941738119849076

Brett, B. L., Kerr, Z. Y., Chandran, A., Walton, S., Aggarwal, N. T., Gifford, K., Mannix, R., DeFreese, J. D., Echemendia, R. J., Guskiewicz, K. M., Meehan, W. P., 3rd., & McCrea, M. A. (2023). A dominance analysis of subjective cognitive complaint comorbidities in former professional football players with and without mild cognitive impairment. *Journal of the International Neuropsychological Society: JINS*, *29*(6), 582–593. https://doi.org/10.1017/s135561772200056x

Brett, B. L., Nader, A. M., Kerr, Z. Y., Chandran, A., Walton, S. R., DeFreese, J. D., Guskiewicz, K. M., & McCrea, M. (2022). Disparate associations of years of football participation and a metric of head impact exposure with neurobehavioral outcomes in former collegiate football players. *Journal of the International Neuropsychological Society: JINS*, *28*(1), 22–34. https://doi.org/10.1017/S1355617721000047

Bryant, A. M., Kerr, Z. Y., Walton, S. R., Barr, W. B., Guskiewicz, K. M., McCrea, M. A., & Brett, B. L. (2023). Investigating the association between subjective and objective performance-based cognitive function among former collegiate football players. *The Clinical Neuropsychologist*, *37*(3), 595–616. https://doi.org/10.1080/13854046.2022.2083021

Caccese, J. B., Bodt, B. A., Iverson, G. L., Kaminski, T. W., Bryk, K., Oldham, J., Broglio, S. P., McCrea, M., McAllister, T., & Buckley, T. A. (2020). Estimated age of first exposure to contact sports and neurocognitive, psychological, and physical outcomes in healthy NCAA collegiate athletes: A cohort study. *Sports Medicine (Auckland, N.Z.)*, *50*(7), 1377–1392. https://doi.org/10.1007/s40279-020-01261-4

Caccese, J. B., DeWolf, R. M., Kaminski, T. W., Broglio, S. P., McAllister, T. W., McCrea, M., & Buckley, T. A. (2019). Estimated age of first exposure to american football and neurocognitive performance amongst NCAA male student-athletes: A cohort study. *Sports Medicine (Auckland, N.Z.)*, *49*(3), 477–487. https://doi.org/10.1007/s40279-019-01069-x

Caccese, J. B., Iverson, G. L., Cameron, K. L., Houston, M. N., McGinty, G. T., Jackson, J. C., O'Donnell, P., Pasquina, P. F., Broglio, S. P., McCrea, M., McAllister, T., & Buckley, T. A. (2020). Estimated age of first exposure to contact sports is not associated with greater symptoms or worse cognitive functioning in male U.S. service academy athletes. *Journal of Neurotrauma*, *37*(2), 334–339. https://doi.org/10.1089/neu.2019.6571

Carlew, A. R., Fatima, H., Livingstone, J. R., Reese, C., Lacritz, L., Pendergrass, C., Bailey, K. C., Presley, C., Mokhtari, B., & Cullum, C. M. (2020). Cognitive assessment via telephone: A scoping review of instruments. *Archives of Clinical Neuropsychology: The Official Journal of the National Academy of Neuropsychologists*, *35*(8), 1215–1233. https://doi.org/10.1093/arclin/acaa096

Carlozzi, N. E., Boileau, N. R., Roché, M. W., Ready, R. E., Perlmutter, J. S., Chou, K. L., Barton, S. K., McCormack, M. K., Stout, J. C., Cella, D., Miner, J. A., & Paulsen, J. S. (2020). Responsiveness to change over time and test-retest reliability of the PROMIS and Neuro-QoL mental health measures in persons with Huntington disease (HD). *Quality of Life Research: An International Journal of Quality of Life Aspects of Treatment, Care and Rehabilitation*, *29*(12), 3419–3439. https://doi.org/10.1007/s11136-020-02596-1

Cella, D., Lai, J.-S., Nowinski, C. J., Victorson, D., Peterman, A., Miller, D., Bethoux, F., Heinemann, A., Rubin, S., Cavazos, J. E., Reder, A. T., Sufit, R., Simuni, T., Holmes, G. L., Siderowf, A., Wojna, V., Bode, R., McKinney, N., Podrabsky, T., … Moy, C. (2012). Neuro-QOL: Brief measures of health-related quality of life for clinical research in neurology. *Neurology*, *78*(23), 1860–1867. https://doi.org/10.1212/WNL.0b013e318258f744

Deshpande, S. K., Hasegawa, R. B., Rabinowitz, A. R., Whyte, J., Roan, C. L., Tabatabaei, A., Baiocchi, M., Karlawish, J. H., Master, C. L., & Small, D. S. (2017). Association of playing high school football with cognition and mental health later in life. *JAMA Neurology*, *74*(8), 909–918. https://doi.org/10.1001/jamaneurol.2017.1317

Eime, R. M., Young, J. A., Harvey, J. T., Charity, M. J., & Payne, W. R. (2013). A systematic review of the psychological and social benefits of participation in sport for children and adolescents: Informing development of a conceptual model of health through sport. *International*

Journal of Behavioral Nutrition and Physical Activity, *10*(1), 98. https://doi.org/10.1186/1479-5868-10-98

Ferris, S. H., Aisen, P. S., Cummings, J., Galasko, D., Salmon, D. P., Schneider, L., Sano, M., Whitehouse, P. J., Edland, S., & Thal, L. J. (2006). ADCS prevention instrument project: Overview and initial results. *Alzheimer Disease and Associated Disorders*, *20*(4 Suppl 3), S109–S123. https://doi.org/10.1097/01.wad.0000213870.40300.21

Fields, L., Didehbani, N., Hart, J., & Cullum, C. M. (2019). No linear association between number of concussions or years played and cognitive outcomes in retired NFL players. *Archives of Clinical Neuropsychology*, *35*(3), 233–239. https://doi.org/10.1093/arclin/acz008

Gershon, R. C., Lai, J. S., Bode, R., Choi, S., Moy, C., Bleck, T., Miller, D., Peterman, A., & Cella, D. (2012). Neuro-QOL: Quality of life item banks for adults with neurological disorders: Item development and calibrations based upon clinical and general population testing. *Quality of Life Research: An International Journal of Quality of Life Aspects of Treatment, Care and Rehabilitation*, *21*(3), 475–486. https://doi.org/10.1007/s11136-011-9958-8

Guskiewicz, K. M., Marshall, S. W., Bailes, J., McCrea, M., Harding, H. P., Jr., Matthews, A., Mihalik, J. R., & Cantu, R. C. (2007). Recurrent concussion and risk of depression in retired professional football players. *Medicine and Science in Sports and Exercise*, *39*(6), 903–909. https://doi.org/10.1249/mss.0b013e3180383da5

Iverson, G. L., Caccese, J. B., Merz, Z. C., Buttner, F., & Terry, D. P. (2021). Age of first exposure to football is not associated with later-in-life cognitive or mental health problems. *Frontiers in Neurology*, *12*, 727089. https://doi.org/10.3389/fneur.2021.647314

Iverson, G. L., Castellani, R. J., Cassidy, J. D., Schneider, G. M., Schneider, K. J., Echemendia, R. J., Bailes, J. E., Hayden, K. A., Koerte, I. K., Manley, G. T., McNamee, M., Patricios, J. S., Tator, C. H., Cantu, R. C., & Dvorak, J. (2023). Examining later-in-life health risks associated with sport-related concussion and repetitive head impacts: A systematic review of case-control and cohort studies. *British Journal of Sports Medicine*, *57*(12), 810–821. https://doi.org/10.1136/bjsports-2023-106890

Iverson, G. L., Merz, Z. C., & Terry, D. P. (2021). High-school football and midlife brain health problems. *Clinical Journal of Sport Medicine: Official Journal of the Canadian Academy of Sport Medicine*, *32*(2), 86–94. https://doi.org/10.1097/jsm.0000000000000898

Iverson, G. L., Terry, D. P., Caccese, J. B., Büttner, F., & Merz, Z. C. (2021). Age of first exposure to football is not associated with midlife brain health problems. *Journal of Neurotrauma*, *38*(5), 538–545. https://doi.org/10.1089/neu.2020.7041

Kerr, Z. Y., Littleton, A. C., Cox, L. M., DeFreese, J. D., Varangis, E., Lynall, R. C., Schmidt, J. D., Marshall, S. W., & Guskiewicz, K. M. (2015). Estimating contact exposure in football using the head impact exposure estimate. *Journal of Neurotrauma*, *32*(14), 1083–1089. https://doi.org/10.1089/neu.2014.3666

Kroenke, K., Spitzer, R. L., & Williams, J. B. (2001). The PHQ-9: Validity of a brief depression severity measure. *Journal of General Internal Medicine*, *16*(9), 606–613. https://doi.org/10.1046/j.1525-1497.2001.016009606.x

Kroenke, K., Strine, T. W., Spitzer, R. L., Williams, J. B., Berry, J. T., & Mokdad, A. H. (2009). The PHQ-8 as a measure of current depression in the general population. *Journal of Affective Disorders*, *114*(1-3), 163–173. https://doi.org/10.1016/j.jad.2008.06.026

Lehman, E. J., Hein, M. J., Baron, S. L., & Gersic, C. M. (2012). Neurodegenerative causes of death among retired national football league players. *Neurology*, *79*(19), 1970–1974. https://doi.org/10.1212/WNL.0b013e31826daf50

Li, C., Neugroschl, J., Luo, X., Zhu, C., Aisen, P., Ferris, S., & Sano, M. (2017). The utility of the cognitive function instrument (CFI) to detect cognitive decline in non-demented older adults. *Journal of Alzheimer's Disease: JAD*, *60*(2), 427–437. https://doi.org/10.3233/jad-161294

Malm, C., Jakobsson, J., & Isaksson, A. (2019). Physical activity and sports-real health benefits: A review with insight into the public health of Sweden. *Sports (Basel)*, *7*(5), 127. https://doi.org/10.3390/sports7050127

Mandolesi, L., Polverino, A., Montuori, S., Foti, F., Ferraioli, G., Sorrentino, P., & Sorrentino, G. (2018). Effects of physical exercise on cognitive functioning and wellbeing: Biological and psychological benefits. *Frontiers in Psychology*, *9*, 509. https://doi.org/10.3389/fpsyg.2018.00509

McAllister, T., & McCrea, M. (2017). Long-term cognitive and neuropsychiatric consequences of repetitive concussion and head-impact exposure. *Journal of Athletic Training*, *52*(3), 309–317. https://doi.org/10.4085/1062-6050-52.1.14

Mez, J., Daneshvar, D. H., Abdolmohammadi, B., Chua, A. S., Alosco, M. L., Kiernan, P. T., Evers, L., Marshall, L., Martin, B. M., Palmisano, J. N., Nowinski, C. J., Mahar, I., Cherry, J. D., Alvarez, V. E., Dwyer, B., Huber, B. R., Stein, T. D., Goldstein, L. E., Katz, D. I., … McKee, A. C. (2020). Duration of American football play and chronic traumatic encephalopathy. *Annals of Neurology*. *87*(1), 116–131. https://doi.org/10.1002/ana.25611

Montenigro, P. H., Alosco, M. L., Martin, B. M., Daneshvar, D. H., Mez, J., Chaisson, C. E., Nowinski, C. J., Au, R., McKee, A. C., Cantu, R. C., McClean, M. D., Stern, R. A., & Tripodis, Y. (2017). Cumulative head impact exposure predicts later-life depression, apathy, executive dysfunction, and cognitive impairment in former high school and college football players. *Journal of Neurotrauma*, *34*(2), 328–340. https://doi.org/10.1089/neu.2016.4413

Randolph, C., Karantzoulis, S., & Guskiewicz, K. (2013). Prevalence and characterization of mild cognitive impairment in retired national football league players. *Journal of the International Neuropsychological Society*, *19*(8), 873–880. https://doi.org/10.1017/S1355617713000805

Reid, M., Parkinson, L., Gibson, R., Schofield, P., D'Este, C., Attia, J., Tavener, M., & Byles, J. (2012). Memory Complaint Questionnaire performed poorly as screening tool: Validation against psychometric tests and affective

measures. *Journal of Clinical Epidemiology*, *65*(2), 199–205. https://doi.org/10.1016/j.jclinepi.2011.06.006

Schaffert, J., Didehbani, N., LoBue, C., Hart, J., Jr., Motes, M., Rossetti, H., Wilmoth, K., Goette, W., Lacritz, L., & Cullum, C. M. (2022). Neurocognitive outcomes of older National Football League retirees. *Brain Injury*, *36*(12-14), 1364–1371. https://doi.org/10.1080/02699052.2022.2143567

Schaffert, J., LoBue, C., Fields, L., Wilmoth, K., Didehbani, N., Hart, J., Jr., & Cullum, C. M. (2019). Neuropsychological functioning in ageing retired NFL players: A critical review. *International Review of Psychiatry (Abingdon, England)*, *32*(1), 71–88. https://doi.org/10.1080/09540261.2019.1658572

Spitzer, R. L., Kroenke, K., Williams, J. B., & Löwe, B. (2006). A brief measure for assessing generalized anxiety disorder: The GAD-7. *Archives of Internal Medicine*, *166*(10), 1092–1097. https://doi.org/10.1001/archinte.166.10.1092

Stamm, J. M., Bourlas, A. P., Baugh, C. M., Fritts, N. G., Daneshvar, D. H., Martin, B. M., McClean, M. D., Tripodis, Y., & Stern, R. A. (2015). Age of first exposure to football and later-life cognitive impairment in former NFL players. *Neurology*, *84*(11), 1114–1120. https://doi.org/10.1212/wnl.0000000000001358

Stillman, A. M., Madigan, N., Torres, K., Swan, N., & Alexander, M. P. (2020). Subjective cognitive complaints in concussion. *Journal of Neurotrauma*, *37*(2), 305–311. https://doi.org/10.1089/neu.2018.5925

Van Patten, R., & Iverson, G. L. (2023). Examining associations between concussion history, subjectively experienced memory problems, and general health factors in older men. *The Clinical Neuropsychologist*, *37*(1), 119–140. https://doi.org/10.1080/13854046.2021.1991481

Wright, M. J., Woo, E., Birath, J. B., Siders, C. A., Kelly, D. F., Wang, C., Swerdloff, R., Romero, E., Kernan, C., Cantu, R. C., & Guskiewicz, K. (2016). An index predictive of cognitive outcome in retired professional American Football players with a history of sports concussion. *Journal of Clinical and Experimental Neuropsychology*, *38*(5), 561–571. https://doi.org/10.1080/13803395.2016.1139057